

Experiment:- 9

Problem statement:-

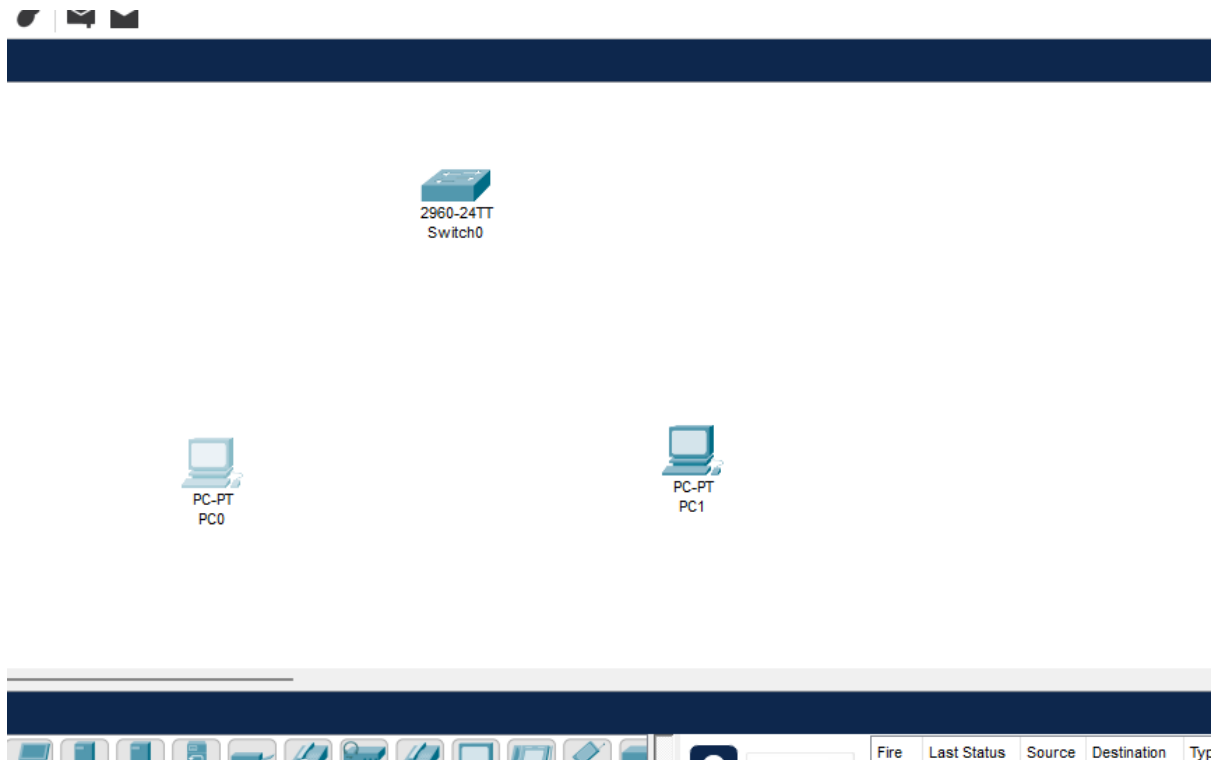
Implement a transport layer simulation to demonstrate process-to-process communication using UDP, TCP, and SCTP protocols. Compare these protocols in terms of connection establishment, data transmission, and congestion control using Cisco Packet Tracer.

Aim:- Transport Layer Simulation: UDP, TCP, and SCTP Comparison

Procedure:-

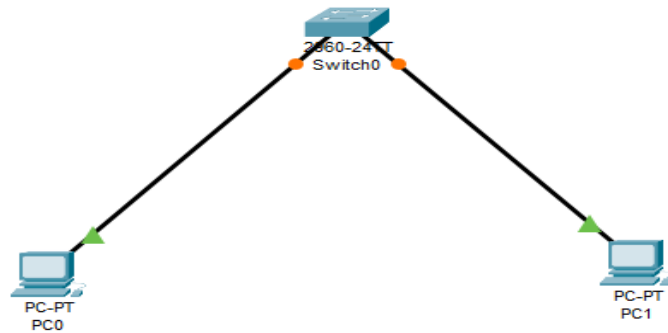
Step 1: Setup the Network Environment

1. Open Cisco Packet Tracer.
2. Create a new project.
3. Add the following devices:
 - Two PCs (PC0 and PC1)
 - One Switch (2960)



4. Connect:

- PC0 → Switch
 - PC1 → Switch
- using Copper Straight-Through cables.



Step 2: Configure IP Addresses

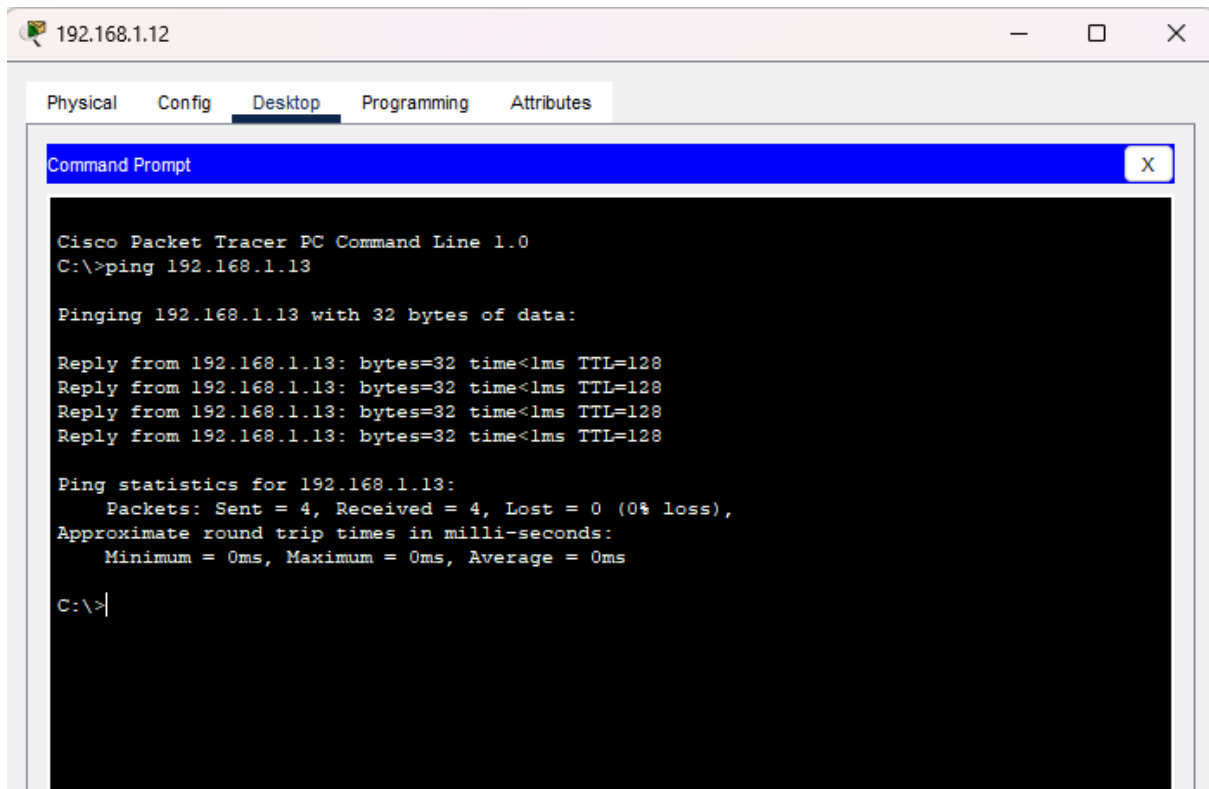
The screenshot shows a window titled '192.168.1.12' with tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, displaying the 'IP Configuration' window for the 'FastEthernet0' interface. The 'IP Configuration' section has two radio buttons: 'DHCP' (unselected) and 'Static' (selected). Below these are fields for 'IPv4 Address' (192.168.1.12), 'Subnet Mask' (255.255.255.0), 'Default Gateway' (0.0.0.0), and 'DNS Server' (0.0.0.0). The 'IPv6 Configuration' section also has two radio buttons: 'Automatic' (unselected) and 'Static' (selected). Below these are fields for 'IPv6 Address' (empty), 'Link Local Address' (FE80::209:7CFF:FE04:8B8), 'Default Gateway' (empty), and 'DNS Server' (empty). The '802.1X' section has a checkbox for 'Use 802.1X Security' (unchecked), an 'Authentication' dropdown menu (set to MD5), and a 'Username' field (empty).

Step 3: Test Connectivity

1. Open Command Prompt on PC0.
2. Type:

```
ping 192.168.1.2
```

3. Successful replies indicate proper network setup.



UDP Communication:-

1. Switch to **Simulation Mode**.
2. Select **Add Simple PDU (envelope icon)**.
3. Click on:
 - o PC0 (source)
 - o PC1 (destination)

The network topology consists of a central switch (Switch0) connected to three PCs (PC-PT). The PCs have IP addresses 192.168.10.1, 192.168.10.2, and 192.168.10.3. The simulation panel on the right shows the event list and play controls.

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
☐	0.000	--

Reset Simulation ☒ Constant Delay Captured to: 0.000 s

Play Controls

If last event, capture then forward (Alt + C)

Event List Filters - Visible Events
DHCPv6, DNS, EIGRPv6, HSRPv6, ICMP, ICMPv6, NDP, OSPFv6, RIPv6, TFTP, UDP

Edit Filters Show All/None

Event List Realtime Simulation

The network topology is the same as the previous screenshot. The simulation panel on the right shows the event list and play controls after a successful UDP communication.

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
☐	0.000	--
☐	0.001	192.168.10.1
☐	0.002	Switch0
☐	0.003	192.168.10.3
☒	0.004	Switch0

Reset Simulation ☒ Constant Delay Captured to: 300.344 s

Play Controls

Event List Filters - Visible Events
DHCPv6, DNS, EIGRPv6, HSRPv6, ICMP, ICMPv6, NDP, OSPFv6, RIPv6, TFTP, UDP

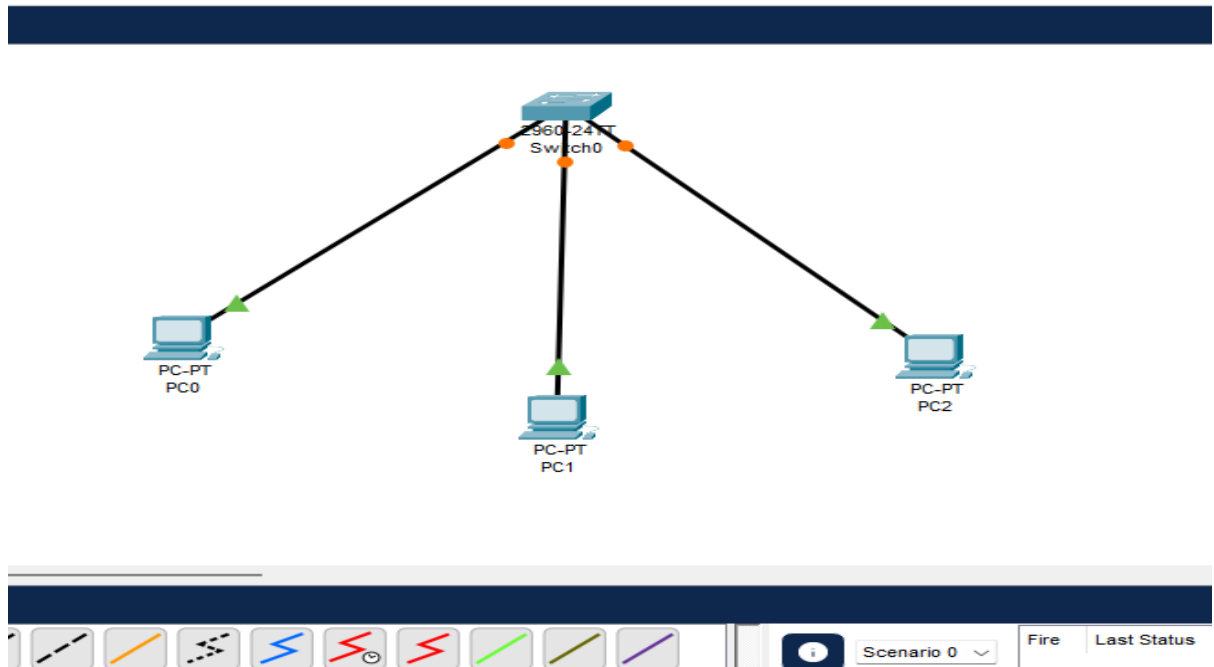
Edit Filters Show All/None

Event List Realtime Simulation

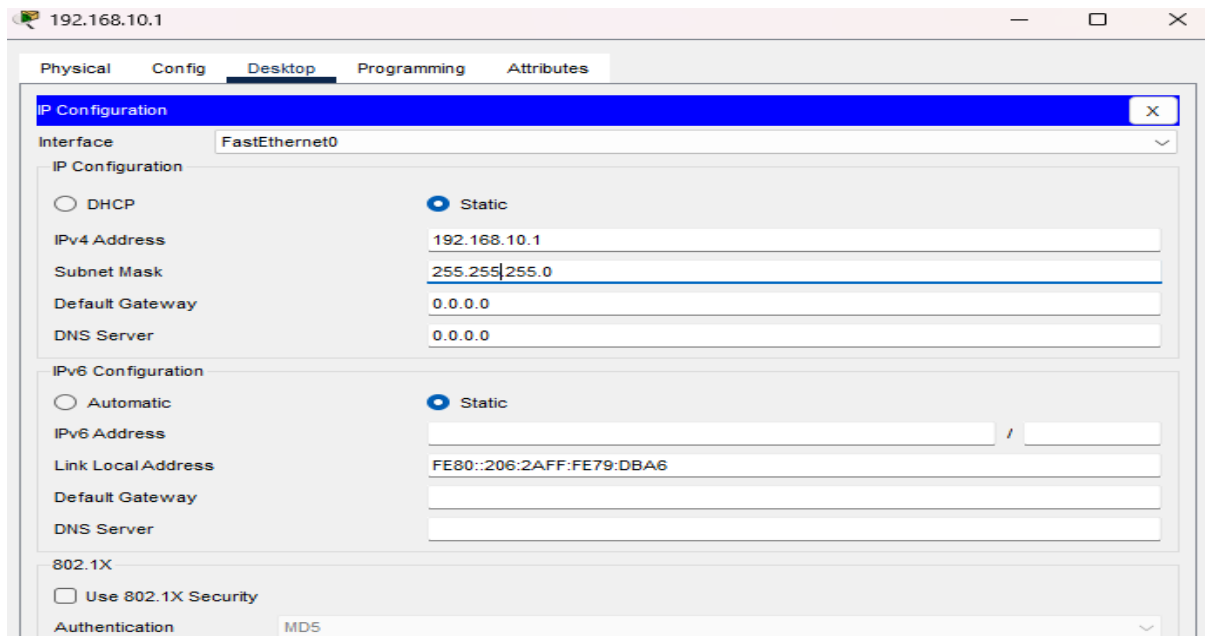
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
☒	Successful	192.1...	192.168.10.3	ICMP	Green	0.000	N	0	(edit)

Simulate TCP Communication:-

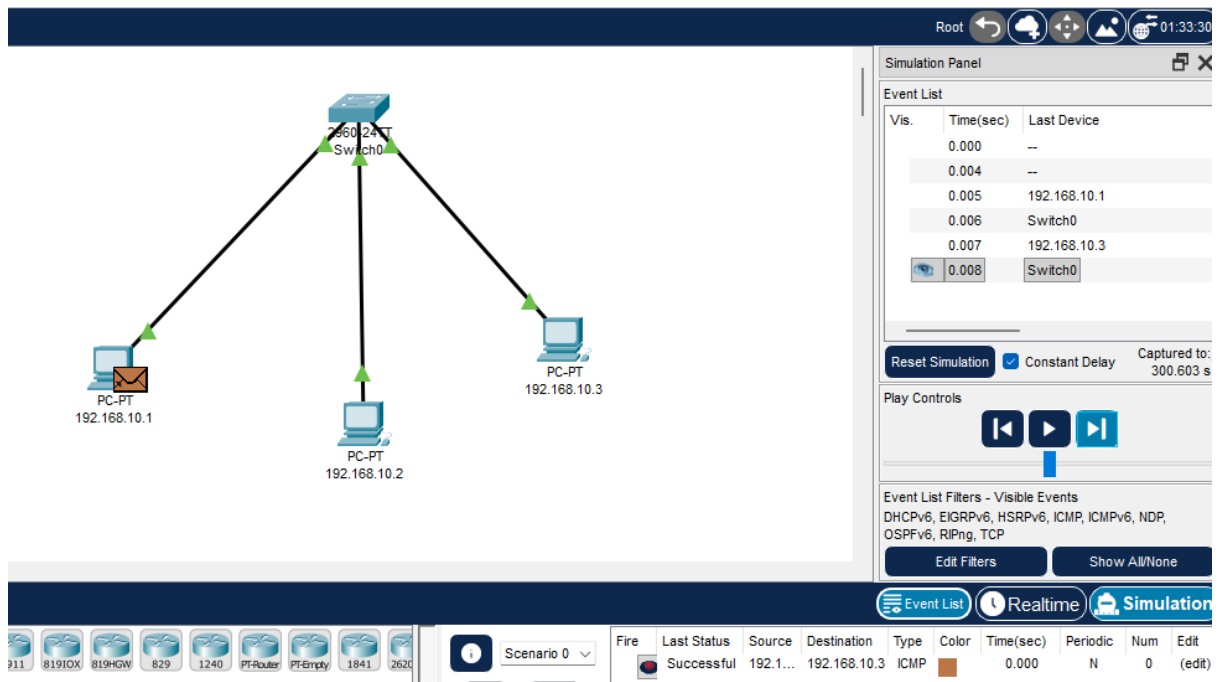
1. Open **Cisco Packet Tracer** and create a new project.
2. Add 2 PCs and 1 Switch, then connect them using **copper straight-through cable**



3. Assign IP addresses:



4. Observe packets in Simulation Mode and verify communication is successful



TCP (Transmission Control Protocol)

- Connection-oriented protocol (connection established before transmission)
- Reliable data transfer (guarantees delivery)
- Maintains correct order of data
- Uses acknowledgments (ACK)
- Retransmits lost packets

UDP (User Datagram Protocol)

- Connectionless protocol (no connection setup)
- Unreliable data transfer (no guarantee of delivery)
- Does not maintain order of data
- No acknowledgments
- No retransmission of lost packets
- No flow control
- No congestion control

